Mohammadjavad Mehditabar

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Summary

Machine Learning Engineer with 5 years of experience across applied R&D and data science, specializing in performance optimization of LLMs and Agentic AI. Passionate about building innovative, practical, and fundamental solutions, I delve into system mechanics to debug root causes and redesign architectures rather than simply applying existing techniques. My work spans building end-to-end ML pipelines, benchmarking LLMs on cost-latency-accuracy trade-offs, profiling hardware-level compute bottlenecks, aligning models, and developing trade-off-aware agentic workflows to build highly efficient and scalable AI systems.

Technical Skills

Programming: Python, C#, Javascript, SQL
Frameworks & Tools: Git, Huggingface, Pytorch, Transformers, Pandas, Numpy, Docker, vLLM, LangChain, , CUDA

Experience

Graduate Research Assistant — Jan. 2025 – Present
SMART Lab, Dalhousie University (Advisor: Prof. Tushar Sharma) — *Nova Scotia, Canada*

- **LLMs Performance Benchmarking** []:
  - Developed two novel Multi-Criteria Decision Making (MCDM) methods to robustly evaluate LLM trade-offs across accuracy, energy consumption, and performance metrics (e.g., throughput and Time To First Token).
  - Profiled quantized and standard language models across Hugging Face, Ollama, and vLLM, analyzing fundamental CUDA compute and memory bottlenecks during prefill and decode phases.
  - Investigated advanced serving techniques—including continuous batching, speculative decoding, and optimal batch sizing—using hardware insights to maximize GPU utilization (MFU) for efficient multi-user deployment.
- **Code Inefficiencies Detector** []:
  - Created a taxonomy of inefficiency patterns in code and curated a high-quality dataset of 3,000 efficient/inefficient code pairs, profiling time, energy, and memory consumption along with reasoning traces to enable downstream RL training.
  - Developed an agentic pipeline using a frontier LLM with dynamic context retrieval, achieving 94% accuracy in identifying inefficiencies within arbitrary code snippets.
- **Meta-Data Aware Movie Recommender []:**
  - Architected an end-to-end GPT-based recommender system fusing textual metadata with user interaction sequences via multi-task learning and LoRA fine-tuning.
  - Improved genre-awareness F1-score from 0.63 to 0.86 and boosted overall recommendation accuracy (NDCG) by up to 5% using a custom genre-filtering algorithm and a decoupled ranking module.

R&D Analyst — Sep. 2025 – Dec. 2025
Lab2Market — *Canada*

- Conducted R&D and 80+ user interviews to evaluate technical solutions optimizing the cost-accuracy-latency triangle in enterprise environments.
- Tested architectural hypotheses against real-world SME workflows to design highly efficient, privacy-preserving systems tailored for sensitive data.
- Analyzed feedback to identify frequent operational bottlenecks and translated user challenges into actionable optimizations.

Research Assistant — Jul. 2021 – Jun. 2024
Iran University of Science Technology (Advisor: Prof. Sauleh Eetemadi) — *Tehran, Iran*

- **Consensus on Machine Learning Models**:
  - Improved the predictive accuracy of traditional machine learning and ensemble models by developing a novel, interactive consensus approach applicable to any supervised learning task.
  - Engineered a dynamic matrix for model convergence and implemented three distinct decision-making layers (soft consensus, hard consensus, and trust factor) to consistently outperform standard baselines.
- **Long Corpus Personality Detection** []:
  - Proposed a novel approach to classify long documents of arbitrary length by leveraging two consecutive BERT models (Hierarchical BERT) to address transformer context limitations.
  - Successfully resolved GPU memory bottlenecks while fine-tuning the full pipeline of weights.

Data Scientist — Jun. 2022 – Jan. 2023
Dadmatech — *Tehran, Iran*

- Engineered end-to-end data pipelines spanning automated web crawling, SQL extraction, and regex-based text mining.
- Developed a custom Django application for efficient user data collection and management.
- Built and optimized a comprehensive suite of machine learning solutions, progressing from classical algorithms to advanced neural networks, transfer learning, and GPT-based language modeling.

Publications

Mehditabar, M., Rajput, S., & Sharma, T. (2026). **"Watts This Smell: A Comprehensive Taxonomy of Software Energy Smells."** IEEE International Conference on Software Maintenance and Evolution (ICSME). IEEE, 2026. [Paper]

Mehditabar, M., Rajput, S., Mastropaolo, A., & Sharma, T. (2026). **"BRACE: Unified Benchmarking of Accuracy and Energy for Code Language Models."** Journal Ahead Workshop (JAWs 2026). [Paper]

Karamdel, H., Ashtiani, M., **Mehditabar, M. J.**, & Bakhshi, F. (2024). **"A consensus-based approach to improve the accuracy of machine learning models."** Evolutionary Intelligence, 1–22. [Paper]

Fatehi, S., Anvarian, Z., Madani, Y., **Mehditabar, M.**, & Eetemadi, S. **"MBTI Personality Prediction Approach on Persian Twitter."** WiNLP at EMNLP 2022 Workshop. [Paper] [Poster]

Honors & Awards

2026 – ICSME Early Acceptance (14 accepted out of 218 submissions)
2025 – Winner, Lab2Market Industrial Proposal
2023 – Ranked 3rd out of 90 students in B.Sc. program
2019 – Ranked in the top 0.4% of 144,000 participants in the National University Entrance Exam

Education

Dalhousie University — Halifax, Nova Scotia, Dalhousie
Master of Computer Science - GPA: A+ — *Jan 2025 – Present*

Iran University of Science and Technology (IUST) — Tehran, Iran
B.Sc. Computer Engineering - GPA: 3.94/4 — *Sep. 2019 – Sep. 2023*